



Beyond Anecdotal Thinking: Deepening Our Understanding for Achieving Quality Goals

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SOFTWARE DEVELOPERS OFTEN build expertise by engaging in new and challenging projects. In this immersive environment, lessons learned are passed from person to person, and there's a tendency toward adopting practices based on experiential and sometimes rather anecdotal evidence.

While there's nothing wrong with this, we can add to and refine our knowledge by empirically evaluating important practices. In this column, we describe how we questioned some widely held beliefs about achieving business agility using service-oriented architectures (SOAs) and challenge you to question and validate your own best practices.

Not to be confused with agile practices such as Extreme Programming (XP) or Scrum, business agility is defined as the ability for an organization to build software systems capable of responding to changing business needs.¹ Organizations seek to deliver software systems that not only satisfy the requirements they know about today but that are sufficiently flexible to accommodate new business opportunities and changing customer needs in the future. Many organizations opt for SOA solutions to in-

crease their financial returns by achieving high degrees of flexibility.

Question the Anecdotes

Motivated by our observations that SOA projects are typically guided by best practices, we decided to empirically explore their effectiveness. We took a software analytics approach to address the question, "Which popular SOA practices concretely contribute to achieving business agility?"

We first documented anecdotal beliefs by asking 20 SOA experts to describe the factors that they believed were essential for delivering business agility. Following an initial brainstorming session, we identified five categories (see Table 1): architectural decisions, business process management (BPM), loose coupling, proactive monitoring and adaptation, and governance.

We worked with these experts to refine each of the categories into a list of factors and then followed the GQM/MEDEA (goal/question/metric metric definition approach) templates and guidelines to define and validate a set of relevant metrics.² From this, we created a questionnaire that included eight questions designed to measure whether



TABLE 1

Business agility factors and associated attributes.

Business agility factor	Description	Associated attributes
Architectural decisions	Tracks major architectural decisions that are known to support flexible environments, such as events, monitoring, analytics, and so on	1.1 Support for rules 1.2 Support for events 1.3 Task automation 1.4 Support for alerts 1.5 Resource allocation 1.6 Support for monitoring 1.7 Support for analytics 1.8 Service architecture
Business process management (BPM)	Tracks the attributes of the BPM component of a given SOA solution; includes use of modeling and simulation	2.1 Use of modeled processes 2.2 Externalized business rules 2.3 Runtime rules housed in engine 2.4 Process orchestration engine 2.5 Monitoring of key performance indicator 2.6 Use of policies
Governance	Tracks the level of governance in a given SOA solution	3.1 Skilled enterprise architect 3.2 Skilled app/SOA architect 3.3 Skilled project manager role 3.4 Skilled release manager role 3.5 Governance documented 3.6 Governance understood 3.7 Governance applied 3.8 Services in repository (discovery) 3.9 Control for requirements changes 3.10 Process for sunseting services 3.11 Percent services governed
Loose coupling	Tracks properties that inject flexibility into a given SOA solution	4.1 Service realization method 4.2 Service access method 4.3 Invokable processes through the Enterprise Service Bus
Proactive monitoring and adaptation	Tracks the properties known to enhance the ability of proactive sensing and adjusting to events	5.1 Reporting analysis 5.2 Requiring service-level agreements (SLAs) 5.3 Measuring SLAs 5.4 Issuing audit reports 5.5 Resource management and utilization 5.6 Proactive monitoring thresholds 5.7 Predictive runtime monitoring 5.8 Historical runtime monitoring 5.9 Life-cycle runtime monitoring

a project achieved business agility following its deployment and more than 70 technical questions that delved into various aspects of the solution and development practices. Answering these technical questions required actually counting, measuring, and evaluating the presence or absence of factors in a real project. Data was painstakingly collected and collated by ar-

chitects and project managers from 32 SOA projects.

From this data, we were able to construct two different instruments. The first was an after-the-fact Business Agility Index (BAI) designed to measure the extent to which a deployed project had achieved business agility. We validated the BAI against 32 projects; it consists of the eight questions in Figure 1. A score of 5 or

higher serves as a strong indicator of achieved business agility.

The second instrument was predictive. An organization could collect a set of metrics from its ongoing (even early-stage) project, and the PBAI (predictive BAI) would predict whether it was headed in the right direction to achieve business agility. The PBAI was developed as a multivariable regression model. Such

1. As a result of deploying this SOA solution, is your organization able to achieve better business outcomes and respond faster to changing customer requirements and demands? (Yes, No)
2. As a result of deploying this SOA solution, is your organization able to address potential ad hoc business situations? (Yes, No)
3. As a result of deploying this SOA solution, is your organization able to adapt better to dynamic business situations once they are identified? (Yes, No)
4. As a result of deploying this SOA solution, is your organization able to create solutions that address market requests and competitors faster and more efficiently? (Yes, No)
5. As a result of deploying this SOA solution, is your organization able to get a better view of its business conditions and react faster to events that have the potential to cause disruption of its business? (Yes, No)
6. As a result of deploying this SOA solution, is your organization able to adapt better to changing situations and enable efficient routing of business needs with minimal interruptions? (Yes, No)
7. As a result of deploying this SOA solution, is your organization able to view trusted and useful data? (Yes, No)
8. As a result of deploying this SOA solution, is your organization able to create services with faster time to the marketplace? (Yes, No)

FIGURE 1. Business Agility Index (BAI) questions about service-oriented architecture (SOA). We score 1 for each “yes” answer; scoring 5 or more serves as a strong indicator of achieved business agility.

TABLE 2

Multiple linear regression model, showing significant factors to be SOA, BPM, and proactive monitoring and adaptation.

	Constant	Architectural decisions	Business process management	Proactive monitoring and adaptation	Loose coupling	Governance
Beta factor	−1.98	.65	.41	.44	.18	.12
Significance	.034	.000**	.027**	.005**	.133*	.309*

* Not significant

** Significant

models are designed to evaluate the contribution of a set of explanatory variables on a dependent variable. In this case, our explanatory variables were the five categories of SOA carefully constructed through the metrics collected from their contributing factors. The dependent variable—that is, the value we wanted to predict, was the attainment of business agility, measured using the BAI. While we can’t go into all the details in this column, know that building such a model requires carefully checking a set of assumptions about linearity and correlations in the data.³

Let the Numbers Speak

Only three of our five hypothesized variables were found to make a sig-

nificant contribution to achieving business agility: architectural decisions, BPM, and proactive monitoring and adaptation. Both governance and loose coupling were rejected. Table 2 reports the overall statistics. These results were interesting, given that our experts had expected good governance practices to contribute to achieving business agility. In hindsight, it makes sense that too much governance might impede change. On the other hand, loose coupling did seem to be a contributing factor, but not at a level that we could show to be statistically significant. An analysis of the model (using the Pearson correlation) showed us that it only explained 63 percent of the variance in business agility, which meant that we had not considered

other significant factors. An obvious one is the impact of inherent project complexity on business agility.

Just building a predictive model isn’t quite enough: we wanted to test its efficacy on our projects, too. Accordingly, we designed a simple experiment following a standard leave-one-out cross-validation design. In this experiment, we repeatedly used 31 projects to build a regression model that in turn we used to predict business agility for the “left out” project. By repeating this process 32 times, we were able to make predictions for each of the projects. Then, by comparing the after-the-fact BAI score for each project versus its predicted BAI, we were able to show that in 87.5 percent of the cases, our model could accurately

predict business agility in advance. Figure 2 shows the results.

Achieving Qualities

Stakeholders' quality concerns come in many different shapes and sizes, and it's an important aspect of the requirements process to understand whether such concerns are achievable or not. Business agility is just one example of a quality concern that, when relevant, must be carefully engineered into the delivered solution. However, there are many others. For example, to build a secure system, we might call in security experts who weigh up many different factors to find the right cost/benefit balance for protecting assets. The same is true for a broad swathe of other quality concerns, such as reliability, performance, maintainability, and scalability.

While anecdotal evidence for achieving such quality concerns is clearly an important ingredient in bridging the gap between requirements and software development, our experiences showed that it's also possible—and advisable—to test and empirically validate the practices we adopt. As software engineers, we should lead the way in thinking about and challenging beliefs. Instead of jumping on various process bandwagons, as we're sometimes apt to do, we can ask deeper questions about whether various solutions really deliver what they promise. In our case, we had initially taken for granted the "fact" that good governance would increase business agility, while in practice, the numbers didn't end up supporting this conjecture. Of course, this doesn't mean we have to throw out SOA governance, but it does mean that we should understand its potentially retarding effect on achieving business agility.

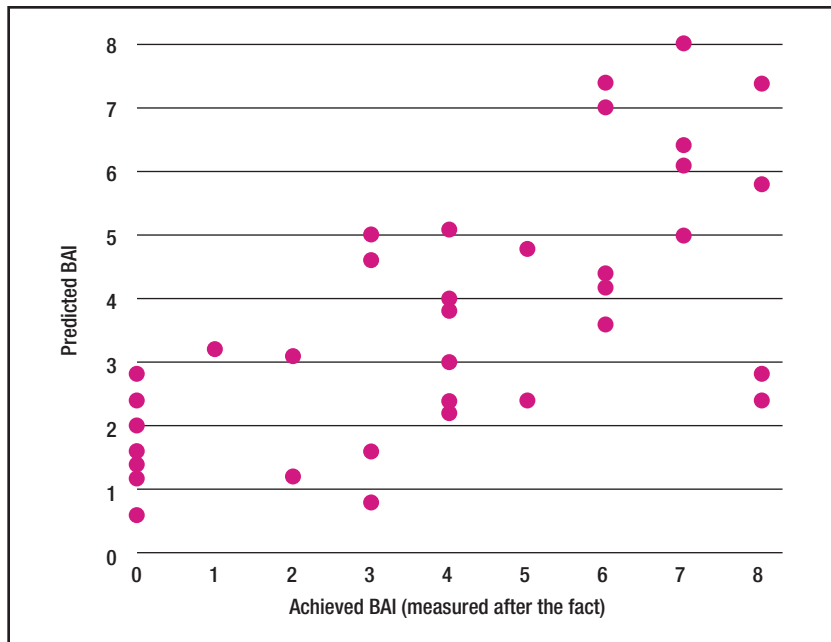


FIGURE 2. Business Agility Index versus predicted BAI for all projects. By comparing the after-the-fact BAI score for each project versus its predicted BAI, we were able to show that in 87.5 percent of the cases, our model could accurately predict business agility in advance.

Software requirements and development practices come and go. At the time we conducted this study, everyone was talking about SOA and business agility. Today, conversations typically focus around big data, social computing, and predictive analytics. However, our underlying message remains the same. Whatever the current trend, we challenge you to think about practices that you might take for granted in your development process. Put on your engineering hat, ask deeper questions, and take the time to empirically question your software engineering beliefs. 

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